

Equivariant Systems Theory and Observer Design for Autonomous Systems

R. Mahony J. Trumpf T. Hamel



Australian
National
University



Module M17, EECI International Graduate School on Control
Zaragoza, 22 June – 26 June 2026

Outline

- 1 Observers
- 2 Gradient observer design
- 3 EqF design
- 4 Conclusion

Outline

- 1 Observers
- 2 Gradient observer design
- 3 EqF design
- 4 Conclusion

Observer architecture

Recall the structure of the lifted system on the Lie group $\mathbf{G}$.

$$\begin{aligned}\dot{X} &= F_{x_0}(X, v) = dL_X \Lambda(\phi_X(x_0), v) \\ y_i &= \rho^i(X, \dot{y}_i) = \rho^i(X, h(x_0))\end{aligned}$$

We propose to use the observer architecture

$$\begin{aligned}\dot{\hat{X}} &= dL_{\hat{X}} \Lambda(\phi_{\hat{X}}(x_0), v) - dR_{\hat{X}} \Delta(\hat{X}, y_1, \dots, y_p) \\ \hat{x}(t) &= \phi_{x_0}(\hat{X}(t))\end{aligned}$$

with initial condition $\hat{X}(0) \in \mathbf{G}$, typically $\hat{X}(0) = I$, and x_0 our best guess for $x(0)$. The correction term Δ remains to be designed.

Equivariant correction terms

We introduce the notations $\mathcal{Y} = \mathcal{Y}_1 \times \cdots \times \mathcal{Y}_p$, $y = (y_1, \dots, y_p)$ and $\dot{y} = (\dot{y}_1, \dots, \dot{y}_p)$. We can also define the product action

$$\rho: \mathbf{G} \times \mathcal{Y} \rightarrow \mathcal{Y}, (S, (y_1, \dots, y_p)) \mapsto (\rho^1(S, y_1), \dots, \rho^p(S, y_p))$$

An *equivariant correction term* is a correction term $\Delta \in \mathfrak{g}$ with

$$\Delta_{\dot{y}}(R_S \hat{X}, \rho(S, y)) = \Delta_{\dot{y}}(\hat{X}, y)$$

for all $S \in \mathbf{G}$ and $y \in \mathcal{Y}$. Note how we have made the parametric dependence of Δ on the reference measurements $\dot{y}$ explicit in the notation.

Having Δ depend explicitly on time or velocity yields no additional advantage.

Observer problem

Kinematic system:

$$\begin{aligned}\dot{x} &= f(x, v) \\ y_i &= h_i(x)\end{aligned}$$

Lifted system:

$$\begin{aligned}\dot{X} &= dL_X \Lambda(\phi_X(x_0), v) \\ y_i &= \rho^i(X, \hat{y}_i)\end{aligned}$$

Observer:

$$\begin{aligned}\dot{\hat{X}} &= dL_{\hat{X}} \Lambda(\phi_{\hat{X}}(x_0), v) - dR_{\hat{X}} \Delta_{\hat{y}}(\hat{X}, y) \\ \hat{x}(t) &= \phi_{x_0}(\hat{X}(t))\end{aligned}$$

The goal: Find a correction term Δ such that some well chosen error variable has “nice” dynamics.

Outline

- 1 Observers
- 2 Gradient observer design
- 3 EqF design
- 4 Conclusion

Equivariance and autonomy

A type I system is a kinematic system with a lifted system of the form

$$\dot{X} = dL_X \Lambda(v).$$

Theorem: Consider a type I system and the observer as above. Then the dynamics of the canonical type I error $E_I(\hat{X}, X) = \hat{X}X^{-1}$ is autonomous if and only if the correction term $\Delta_{\hat{y}}(\hat{X}, y)$ is equivariant.

We can explicitly compute the error dynamics in this case.

$$\dot{E}_I = -dR_{E_I} \Delta_{\hat{y}}(E_I, \hat{y}).$$

(For type II systems, the appropriate error variable is the canonical type II error $E_I(\hat{X}, X) = X^{-1}\hat{X}$).

The specific form of these error dynamics invites a **gradient based correction term design**.

Invariant cost functions

We start by defining *invariant cost functions* on the output spaces $\mathcal{Y}_i$.

$$\ell_{\hat{y}_i}^i: \mathbf{G} \times \mathcal{Y}_i \rightarrow \mathbb{R}^+, (\hat{X}, y_i) \mapsto \ell_{\hat{y}_i}^i(\hat{X}, y_i)$$

where

$$\ell_{\hat{y}_i}^i(R_S \hat{X}, \rho^i(S, y_i)) = \ell_{\hat{y}_i}^i(\hat{X}, y_i)$$

for all $S \in \mathbf{G}$. Then $\ell_{\hat{y}_i}^i(\hat{X}, y_i) = \ell_{\hat{y}_i}^i(E_1, \hat{y}_i) = \ell_{\hat{y}_i}^i(I, e_i^!)$ where $e_i^! = \rho^i(E_1^{-1}, \hat{y}_i)$.

We require $(I, \hat{y}_i)$ to be a global minimum of $\ell_{\hat{y}_i}^i$. Moreover, we require that the cost is *non-degenerate* in the sense that $\text{Hess}_2 \ell_{\hat{y}_i}^i(I, \hat{y}_i) > 0$.

Constructing invariant cost functions

Starting from a single variable cost function

$$f_{\mathring{y}_i}^i: \mathcal{Y}_i \rightarrow \mathbb{R}^+$$

with a global minimum at $y_i = \mathring{y}_i$, the construction

$$\ell_{\mathring{y}_i}^i(\hat{X}, y_i) := f_{\mathring{y}_i}^i(\rho^i(\hat{X}^{-1}, y_i)) = f_{\mathring{y}_i}^i(e_i^!)$$

yields an invariant cost function with global minimum at $(I, \mathring{y}_i)$
 that is non-degenerate if $f_{\mathring{y}_i}^i$ is non-degenerate at $\mathring{y}_i$.

Aggregate cost

The aggregate cost across all outputs is

$$\ell_{\dot{y}}: \mathbf{G} \times \mathcal{Y} \rightarrow \mathbb{R}^+, \ell_{\dot{y}}(\hat{X}, y) = \sum_{i=1}^p \ell_{\dot{y}_i}^i(\hat{X}, y_i) = \ell_{\dot{y}}(E_1, \dot{y}) = \ell_{\dot{y}}(I, e^1)$$

Lemma: $\ell_{\dot{y}}$ is invariant and $\ell_{\dot{y}}(\cdot, \dot{y}): \mathbf{G} \rightarrow \mathbb{R}^+$ has a global minimum at $E_1 = I$. It is non-degenerate if all the $\ell_{\dot{y}_i}^i$ are non-degenerate and

$$\bigcap_{i=1}^p \text{stab}_{\rho^i}(\dot{y}_i) = \{I\}$$

For type I systems, the aggregate cost $\ell_{\dot{y}}(\cdot, \dot{y}): \mathbf{G} \rightarrow \mathbb{R}^+$ can be directly used for observer design.

Gradient correction term

We define the *gradient correction term*

$$\Delta_{\hat{y}}(\hat{X}, y) = k dR_{\hat{X}^{-1}}(I) \operatorname{grad}_{\hat{X}} \ell_{\hat{y}}(\hat{X}, y)$$

where the gradient is taken with respect to a right invariant Riemannian metric on $\mathbf{G}$. The latter is constructed as $\langle dR_S V, dR_S W \rangle_S = \langle V, W \rangle$ where $V, W \in \mathfrak{g}$ and $\langle \cdot, \cdot \rangle$ is an inner product on $\mathfrak{g}$.

$$\begin{aligned} \frac{d}{dt} \ell_{\hat{y}}(E_1, \hat{y}) &= \sum_{i=1}^p d_2 \ell_{\hat{y}_i}^i(I, e_i^1) d\rho_{e_i^1}^i(I) [\Delta] \\ &= -k \|\operatorname{grad}_{\hat{X}} \ell_{\hat{y}}(E_1, \hat{y})\|^2 \end{aligned}$$

Observer error convergence

Theorem: Consider a type I kinematic system and invariant cost functions $\ell_{\dot{y}_i}^i : \mathbf{G} \times \mathcal{Y}_i \rightarrow \mathbb{R}^+$ with

$$\bigcap_{i=1}^p \text{stab}_{\rho^i}(\dot{y}_i) = \{I\}$$

Define the observer

$$\begin{aligned}\dot{\hat{X}} &= dL_{\hat{X}}\Lambda(v) - k \text{grad}_{\hat{X}} \ell_{\dot{y}}(\hat{X}, y) \\ \hat{x}(t) &= \phi_{x_0}(\hat{X}(t))\end{aligned}$$

Then there exists a basin of attraction $I \in \mathcal{B} \subset \mathbf{G}$ such that $E_I(0) \in \mathcal{B}$ implies $E_I(t) \rightarrow I$ and $\hat{X}(t) \rightarrow X(t)$.
 If $\phi_{x_0}(X(0)) = x(0)$ then $\hat{x}(t) \rightarrow x(t)$.

Outline

- 1 Observers
- 2 Gradient observer design
- 3 EqF design**
- 4 Conclusion

Error dynamics revisited

For a kinematic system and observer as above, the *state error*

$$e = e(\hat{X}, x) := \phi(\hat{X}^{-1}, x)$$

has dynamics

$$\dot{e} = d\phi_e \operatorname{Ad}_{\hat{X}} (\Lambda(\phi_{\hat{X}}(e), v) - \Lambda(\phi_{\hat{X}}(x_0), v)) - d\phi_e \Delta_{\hat{y}}(\hat{X}, y).$$

For an *equivariant* kinematic system this simplifies to

$$\dot{e} = d\phi_e (\Lambda(e, v_t^0) - \Lambda(x_0, v_t^0)) - d\phi_e \Delta_{\hat{y}}(\hat{X}, y)$$

where $v_t^0 := \psi_{\hat{X}^{-1}}(v(t))$.

local coordinates and linearisation

Choose a local coordinate chart $\epsilon: \mathcal{U} \rightarrow \mathbb{R}^m$, $e \mapsto \epsilon(e)$ with $x_0 \in \mathcal{U} \subset \mathcal{M}$ and $\epsilon(x_0) = 0$.

Choose a local coordinate chart $\delta: \mathcal{W} \rightarrow \mathbb{R}^{n_1 + \dots + n_p}$ of the output (product) manifold $\mathcal{N}$ such that both $y \in \mathcal{W}$ and $\hat{y} \in \mathcal{W}$. Define $\tilde{\delta} := \delta(y) - \delta(\hat{y})$.

The linearised error dynamics around $e = x_0$ ($\epsilon = 0$) in these coordinates then have the form

$$\dot{\epsilon} = A_t \epsilon + D \Delta$$

$$\tilde{\delta} = C_t \epsilon$$

where we might have to *switch* the coordinates δ whenever y or $\hat{y}$ moves too much.

local coordinates and linearisation

$$A_t := D_{e|x_0} \varepsilon(e) \cdot D_{E|\text{id}} \phi_{x_0}(E) \cdot D_{e|x_0} \Lambda(e, v_t^0) \cdot D_{\varepsilon|0} \varepsilon^{-1}(\varepsilon),$$

$$C_t := D_{y|y_t} \delta(y) \cdot D_{x|\hat{x}} h(x) \cdot D_{e|x_0} \phi_{\hat{x}}(e) \cdot D_{\varepsilon|0} \varepsilon^{-1}(\varepsilon),$$

$$D := D_{e|x_0} \varepsilon(e) \cdot D_{E|\text{id}} \phi_{x_0}(E)$$

If the output is *equivariant* and allows for a single output manifold chart then $C_t \equiv C$ is *constant*! In this case

$$C = D_{d|y} \delta(d) \cdot D_{e|x_0} h(e) \cdot D_{\varepsilon|0} \varepsilon^{-1}(\varepsilon)$$

where $d = e^l = \rho_{\hat{x}^{-1}}(y)$.

The Equivariant Filter (EqF)

The EqF is based on a Riccati observer design

$$\begin{aligned}\dot{\hat{X}} &= dL_{\hat{X}}\Lambda(\phi_{x_0}(\hat{X}), v) \\ &\quad - dR_{\hat{X}}d\phi_{x_0}^\dagger D_{\varepsilon|0}\varepsilon^{-1}(\varepsilon)\Sigma C^t{}^\top N_t^{-1} \left[\delta(y) - \delta(h(\phi_{x_0}(\hat{X}))) \right]\end{aligned}$$

$$\dot{\Sigma} = A_t \Sigma + \Sigma A_t^\top + M_t - \Sigma C_t^\top N_t^{-1} C_t \Sigma,$$

$$\hat{x}(t) = \phi_{x_0}(\hat{X}(t))$$

with initial conditions $\hat{X}(0) \in \mathbf{G}$ (usually $\hat{X}(0) = I$) and $\Sigma(0) = \Sigma_0$ positive definite, and gain matrices M_t and N_t .

EqF error convergence

Theorem: Consider an equivariant kinematic system with equivariant output and equivariant lift. Let M_t and N_t be positive definite symmetric gain matrices with

$$\lambda I_m < M_t$$

for a constant $\lambda > 0$.

If solutions $(\hat{X}(t), \Sigma(t))$ to the equivariant filter equation exist for $t \rightarrow \infty$ and

$$\sigma_1 I_m < \Sigma(t) < \sigma_2 I_m$$

for constants $\sigma_2 > \sigma_1 > 0$ then the state error $e = \phi(\hat{X}^{-1}, x)$ is uniformly locally exponentially stable in a neighbourhood of $e = x_0$.

Outline

- 1 Observers
- 2 Gradient observer design
- 3 EqF design
- 4 Conclusion

Examples that we covered

Attitude observer: $\dot{R} = R\Omega_{\times}$, Type I system on $\text{SO}(3)$, gradient error dynamics (*beautiful!!!*)

Attitude observer with bias: extensions to that theory with Lyapunov analysis (*cool!!!*)

Homography observer: $\dot{H} = H(\Omega_{\times} + \Gamma)$, *not* Type I (because Γ depends on state information), but we model as Type I with bias (*works!!!*)